

# Pixels to Predictions: Fine-tuning SmolVLM-500M for Scientific Visual Question Answering

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## Abstract

We fine-tune SmolVLM-500M-Instruct (500M parameters) for scientific visual multiple-choice question answering under a 5-million trainable parameter constraint. Across 33 leaderboard submissions, 26+ experiments and ablation studies, and 16 documented bug fixes, we improve from a zero-shot baseline of 0.625 to a final public leaderboard accuracy of **0.889**. Our unified experimentation pipeline first establishes systematic ablation studies over LoRA targets, ranks, learning rates, label masking, resolution, and prompts, achieving a best single-model score of 0.879. We then extend the strongest configuration with DoRA ( $r=8$ ,  $\alpha=16$ , dropout=0.05) targeting all seven projection layers (q\_proj, k\_proj, v\_proj, o\_proj, gate\_proj, up\_proj, down\_proj), self-captioning, metadata enrichment, train+val combined training, and score-average ensembling across 8 diverse configurations. We provide complete hyperparameter specifications for all models and full reproduction steps.

## 1 Introduction

The Pixels to Predictions competition requires building a multimodal model that selects the correct answer from image-text multiple-choice science questions. Each sample contains an image (diagram, chart, map, or photograph), a question, 2–5 answer choices, and optional context (lecture text and hint). Competition constraints are strict: the only allowed base model is SmolVLM-500M-Instruct (500M parameters), trainable parameters are capped at 5 million, only the provided dataset may be used, compute must stay within free-tier resources (Google Colab Free Tier and Kaggle Free Tier), and evaluation runs offline without internet access.

We organize the project as a single, phase-driven experimental campaign. The early phases focus on systematic ablation, exploring LoRA targets, ranks, learning rates, label masking strategies, inference methods, resolution, and prompt content. The later phases build on the strongest findings and introduce DoRA [3], self-captioning, metadata enrichment, train+val combined training, and multi-model score-average ensembling. This unified workflow allowed fixes and insights from each phase to feed directly into subsequent experiments.

**Contributions.** (1) We present controlled ablation studies isolating the effect of LoRA targets, image resolution, prompt content, inference method, epoch count, and label masking, with confounded comparisons explicitly marked. (2) We demonstrate that score-averaged ensembling of diverse single models (different seeds, adapter types, data strategies) provides reliable gains (+1.0pp over best single model). (3) We document 16 technical bugs specific to multimodal LoRA fine-tuning, many of which are non-obvious failure modes.

Our final best submission, `sub_score_average.csv`, achieves **0.889** on the public leaderboard, a +26.4 percentage point improvement over the zero-shot baseline of 0.625.

## 2 Hardware and Environment

All training and inference experiments were conducted using Google Colab Student Subscription environments with NVIDIA A100 40GB GPUs. The software stack was Python 3.12, PyTorch 2.x with CUDA 12.1, Transformers 4.57.1, PEFT 0.18.0, bitsandbytes 0.45.x, Accelerate 1.7.x, and TRL 0.18.1, which was tested but not used in the final pipeline. Google Drive was used for checkpoint persistence

across Colab sessions. Total compute was approximately 120 GPU-hours across all experiments.

### 3 Dataset and Validation Strategy

The competition provides training samples, validation samples, and hidden test samples drawn from ScienceQA. Each sample has an image, a question, 2–5 answer choices (median 3), and optional lecture and hint text. The dataset spans three subjects (natural science, social science, language science) and 26 topic categories.

| Split      | Samples | Usage                    |
|------------|---------|--------------------------|
| Train      | 3,109   | LoRA/DoRA fine-tuning    |
| Validation | 1,048   | Hyperparameter selection |
| Test       | Hidden  | Leaderboard submission   |

Table 1: Dataset splits. 85% of samples include lecture text; 83% include hints.

**Validation protocol.** Early experiments (v1–v5) used a 200-sample random subset for rapid iteration, which proved unreliable—confidence routing improved validation by +0.5pp but failed to transfer to the leaderboard. From Phase 3 onward, we used the full 1,048-sample validation set for model selection and ablation checks. For final Phase 4 models, we trained on the combined train+val set only after hyperparameters had been fixed using the validation split; these final runs were treated as blind leaderboard submissions.

**Hard sample analysis.** Running v2 inference on the full training set identified 541 incorrect predictions (17.4%). Hard samples had longer prompts (1,416 vs. 1,274 chars) and more frequently included lecture text (96% vs. 84%). Fine-tuning on hard samples *degraded* leaderboard score (0.754 vs. 0.768), confirming that these samples are genuinely ambiguous rather than learnable.

**Confidence analysis.** High-confidence predictions (log-prob margin  $\geq 3.0$ ) achieved 84.5% accuracy, while low-confidence (margin  $< 1.0$ ) achieved only 51.5%, approaching random guessing.

### 4 Model Architecture

SmolVLM-500M-Instruct consists of a 93M-parameter SigLIP-B/16 vision encoder, a pixel-shuffle projection layer, and a 360M-parameter SmolLM2-360M language model (total: 502M parameters). The vision encoder is relatively weak compared to the language model, which motivates our self-captioning approach (§8.1). The model was loaded in float16 precision with `device_map="auto"`.

### 5 LoRA and DoRA Configuration

Table 2 presents the complete hyperparameter specification for three key configurations: the Phase 2 v2 model, the strongest single v9 model, and the final ensemble models.

**Why DoRA over LoRA.** Standard LoRA [1] on all 7 projection layers caused overfitting in early ablations at  $\text{LR}=3 \times 10^{-5}$  (§7.1). The v9 model bypassed this with a higher flat LR ( $2 \times 10^{-4}$ ), while later DoRA runs found that DoRA—which decomposes weight updates into magnitude and direction components—provides implicit regularization allowing all 7 targets at the lower, more stable LR. The best individual model (Model E,  $r=8$ ,  $\alpha=16$ , seed 42, no captioning, standard LoRA) achieved 0.879 on the leaderboard.

**Why  $r=8$ ,  $\alpha=16$ .** With 7 target layers,  $r=8$  produces  $\sim 4.8\text{M}$  trainable parameters, near the 5M cap.  $\alpha=16$  (ratio  $\alpha/r = 2$ ) provides moderate scaling. Model D tested  $\alpha=32$  (LB 0.863), underperforming  $\alpha=16$  (Model A, LB 0.871), justifying the  $2 \times$  ratio.

### 6 Inference Pipeline

**Log-likelihood scoring.** For each question, each answer choice is scored independently. The prompt ends with “Answer:” and we compute the log-probability of each candidate letter  $c \in \{A, B, C, \dots\}$  at the first generated position. Multi-form token scoring (“A”, “A”, “A.”, “ A.”) takes the maximum across forms per choice.

**Critical bug: batched inference padding.** With left-padding and batch size  $> 1$ , naively reading logits from position  $[-1]$  reads from a padding position for shorter sequences. Fixing this to read from the actual

| Hyperparameter                                  | Phase 2 v2 (LB 0.768)    | Phase 3 v9 (LB 0.861)    | Final Ensemble (LB 0.889) |
|-------------------------------------------------|--------------------------|--------------------------|---------------------------|
| Adaptation method                               | LoRA                     | LoRA                     | DoRA [3]                  |
| Rank ( $r$ )                                    | 16                       | 8                        | 8                         |
| Alpha ( $\alpha$ )                              | 32                       | 16                       | 16                        |
| LoRA/DoRA dropout ( <code>lora_dropout</code> ) | 0.05                     | 0.05                     | 0.05                      |
| Target modules                                  | q,k,v,o_proj             | all 7 projections        | all 7 projections         |
| Trainable params                                | 1,105,920                | $\sim 4,300,000$         | 4,784,128                 |
| Bias                                            | none                     | none                     | none                      |
| Task type                                       | CAUSAL_LM                | CAUSAL_LM                | CAUSAL_LM                 |
| Batch size                                      | 1                        | 1                        | 1                         |
| Gradient accum.                                 | 16 (eff. 16)             | 16                       | 16 (eff. 16)              |
| Learning rate                                   | $3 \times 10^{-5}$       | $2 \times 10^{-4}$       | $3 \times 10^{-5}$        |
| LR scheduler                                    | Cosine (5% warmup)       | Flat (none)              | Flat (none)               |
| Optimizer                                       | AdamW ( $\lambda=0.01$ ) | AdamW ( $\lambda=0.01$ ) | AdamW ( $\lambda=0.01$ )  |
| Epochs                                          | 2                        | 3                        | 2                         |
| Max grad norm                                   | 1.0                      | 1.0                      | 1.0                       |
| Precision                                       | float16                  | float16                  | float16                   |
| Grad. checkpoint                                | Enabled (non-reentrant)  | Enabled                  | Enabled (non-reentrant)   |
| Seed                                            | 42                       | 42                       | 42, 123, 456, 789         |
| Image res. (train)                              | $224 \times 224$         | Native                   | $224 \times 224$          |
| Image res. (infer.)                             | Native                   | Native                   | Native                    |
| Training target                                 | Answer phrase            | Letter only              | Answer phrase             |
| Training data                                   | Train only (3,109)       | Train+Val (4,157)        | Train+Val (4,157)         |

Table 2: Complete hyperparameters for three key configurations. Target modules: `q_proj`, `k_proj`, `v_proj`, `o_proj` (4 attention) or all 7 including `gate_proj`, `up_proj`, `down_proj` (MLP). The v9 run discovered that flat LR= $2 \times 10^{-4}$  with all 7 targets works (contradicting earlier ablations where MLP targets hurt at LR= $3 \times 10^{-5}$ ), revealing a LR–target interaction. The final ensemble adopted all-7 targets with DoRA regularization at the lower LR.

last non-padding token was one of the most impactful corrections.

**Native image resolution.** Images loaded at original resolution without resizing preserve fine diagram details. We observed +1.0pp from native inference on a 224px-trained model (0.758 $\rightarrow$ 0.768). We adopted native inference for all final submissions.

**Inference method comparison.** Letter log-prob and greedy generation produced identical accuracy (0.670 on val, 1 disagreement in 200 samples). Full log-likelihood scoring (summing log-probs across all answer tokens) was the worst method (0.535), suffering from length bias. Chat template inference degraded to 0.450, confirming the model responds better to raw instructional prompts.

## 7 Ablation Studies

Where possible, ablations change one variable while holding the remaining settings fixed; when an experi-

ment changes multiple variables, we explicitly mark it as a confounded comparison and do not attribute the score change to a single factor.

### 7.1 LoRA Target Modules

**Controlled:** Target layers. **Held constant:**  $r=16$ ,  $\alpha=32$ , LR= $3 \times 10^{-5}$ , 2 epochs, 224px, standard LoRA, dropout=0.05, cosine scheduler with 5% warmup.

**Confound note:** The  $r=4$  (q,v) and  $r=16$  (q,k,v,o) rows differ in both target modules *and* rank/alpha. The 0.722 $\rightarrow$ 0.758 jump conflates these changes. However, subsequent rows consistently show MLP degradation under standard LoRA at LR= $3 \times 10^{-5}$ , isolating the target–LR interaction.

### 7.2 Label Masking Strategy

**Controlled:** Supervision signal. **Held constant:** v2 config ( $r=16$ , q/k/v/o, LR= $3 \times 10^{-5}$ , 2 epochs).

| Targets           | $r/\alpha$   | Params      | LB           |
|-------------------|--------------|-------------|--------------|
| q, v only         | 4/8          | 276K        | 0.722        |
| <b>q, k, v, o</b> | <b>16/32</b> | <b>1.1M</b> | <b>0.758</b> |
| q,k,v,o + MLP     | 8/16         | 4.8M        | 0.754        |
| All 7 ( $r=8$ )   | 8/16         | 4.8M        | 0.730        |
| Text-only (all 7) | 8/16         | 4.3M        | 0.766        |

Table 3: LoRA targets (standard LoRA,  $LR=3\times 10^{-5}$ ). MLP targets hurt at this LR. The v9 run later found that flat  $LR=2\times 10^{-4}$  enables all-7 targets (LB 0.861), and DoRA later achieved the same at  $LR=3\times 10^{-5}$ .

| Strategy                                | LB           |
|-----------------------------------------|--------------|
| Single letter token (“A”)               | 0.722        |
| <b>Answer phrase (“A. choice text”)</b> | <b>0.758</b> |
| Count-from-end masking                  | 0.754        |
| Full sequence (no masking)              | 0.499        |

Table 4: Label masking. Answer-phrase training provides 5–15 tokens of gradient signal vs. 1 for letter-only, yielding +3.6pp. Full-sequence training collapses to noise.

### 7.3 Image Resolution

**Controlled:** Image size at inference. **Held constant:** LoRA v2 model (trained at 224px), full prompt, log-prob scoring.

### 7.4 Prompt Content

**Controlled:** Context fields included. **Held constant:** LoRA v2, 224px, log-prob scoring.

### 7.5 Training Epochs

**Controlled:** Epoch count. **Held constant:** Attention-only,  $LR=3\times 10^{-5}$ , dropout=0.05.

| Resolution                 | Val          | LB           |
|----------------------------|--------------|--------------|
| $224 \times 224$           | 0.670        | 0.758        |
| $336 \times 336$           | 0.655        | —            |
| <b>Native</b>              | <b>0.675</b> | <b>0.768</b> |
| $512 \times 512$ (trained) | —            | 0.762        |

Table 5: Image resolution. Native resolution preserves diagram details that fixed-size resizing destroys, even outperforming a model trained at 512px.

| Prompt Variant                          | Val          |
|-----------------------------------------|--------------|
| <b>Full (lecture + hint + question)</b> | <b>0.670</b> |
| Question + hint only                    | 0.595        |
| Question + lecture only                 | 0.585        |
| Question only                           | 0.565        |

Table 6: Prompt content. Lecture contributes  $\sim 7.5$ pp, hint  $\sim 8.5$ pp. Full context is always best.

| Epochs   | Final Loss   | LB           |
|----------|--------------|--------------|
| 1        | 0.138        | 0.722        |
| <b>2</b> | <b>0.071</b> | <b>0.758</b> |
| 3        | 0.069        | 0.754        |

Table 7: Epochs. 2 is optimal at  $LR=3\times 10^{-5}$ ; epoch 3 overfits (loss decreases but LB degrades). The v9 run used 3 epochs at higher  $LR=2\times 10^{-4}$  without overfitting.

## 8 Phase 4: Advanced Techniques

Building on the ablation results and the v9 discovery that flat LR enables all-7 targets, we introduced several advanced techniques.

### 8.1 Self-Captioning

Given the asymmetry between the weak vision encoder (93M SigLIP) and the stronger language model (360M SmoLM2), we hypothesized that converting visual information to text would leverage the model’s strength. We used the *base* SmoLVLM-500M-Instruct (before fine-tuning) to generate text descriptions of each image using the prompt “Describe this image in detail for a science question.” These captions were prepended as “Image description: {caption}”. This complies with competition rules: only the allowed model and provided data are used.

### 8.2 Metadata Enrichment

We added subject and grade metadata to prompts (e.g., “Subject: natural science | Grade: grade8”), providing domain context that helps disambiguate visually similar questions across subjects.

### 8.3 DoRA (Weight-Decomposed LoRA)

We switched from standard LoRA to DoRA [3], which decomposes weight updates into magnitude

| ID | Method      | Config                                             | LB           |
|----|-------------|----------------------------------------------------|--------------|
| A  | DoRA        | $r=8/\alpha=16$ , cap+meta, seed 42                | 0.871        |
| B  | DoRA        | $r=8/\alpha=16$ , cap+meta, seed 123               | 0.845        |
| C  | DoRA        | $r=8/\alpha=16$ , cap+sol, seed 456                | 0.787        |
| D  | DoRA        | $r=8/\alpha=32$ , cap+meta, seed 789               | 0.863        |
| E  | <b>LoRA</b> | <b><math>r=8/\alpha=16</math>, no cap, seed 42</b> | <b>0.879</b> |
| F  | DoRA        | $r=16/\alpha=32$ , cap, cos, seed 42               | —            |
| G  | DoRA        | $r=8/\alpha=16$ , CoT prompt, seed 777             | —            |
| H  | DoRA        | $r=8/\alpha=16$ , cap, LR=5e-5, seed 333           | —            |

Table 8: Individual model configurations. “cap”=self-captioning, “meta”=subject/grade metadata, “sol”=solution field, “cos”=cosine scheduler. All use all 7 target modules, dropout=0.05, train+val data, native-resolution inference. Models F–H were not submitted individually.

and direction components. This provided implicit regularization that allowed targeting all 7 projection layers at  $r=8$  without the overfitting observed in the earlier ablations (under standard LoRA at  $\text{LR}=3\times 10^{-5}$ ).

#### 8.4 Train+Validation Combined Training

For final models, training on the combined train+val set after fixing hyperparameters yielded a 34% data increase. The v9 run independently confirmed this approach (v9 train+val: LB 0.861).

#### 8.5 Diverse-Configuration Ensemble

Our strongest results came from ensembling models with systematically varied configurations:

**Ensemble method.** Score-average ensembling: for each test sample, log-probability scores from all models are averaged, and the choice with the highest average is selected. This preserves confidence information better than majority voting.

#### Ensemble composition ablation.

**Key insight: Model C hurts.** Model C included the solution field in training, degrading individual accuracy to 0.787. Including C consistently reduced ensemble performance (0.883 with C vs. 0.885 without), confirming that training on solution text introduces noise.

| Ensemble Composition                  | LB           |
|---------------------------------------|--------------|
| A + E (2-model)                       | 0.873        |
| A + B + E (3-model)                   | 0.875        |
| A + B + D + E (4-model, no C)         | 0.885        |
| All 8 models (A–H)                    | 0.887        |
| <b>Score average (optimal subset)</b> | <b>0.889</b> |

Table 9: Ensemble ablation. Adding diverse models helps; dropping weak Model C helps. Score-average over the optimal subset achieves the best result.

## 9 All Submitted Models

Table 10 presents all 33 leaderboard submissions organized by phase. Submission filenames and scores match the leaderboard exactly.

Table 10: All 33 submitted models. Best overall: sub\_score\_average.csv (**0.889**).

| #  | Phase                             | Submission filename                                                       | Key Config / Approach                                                               | Type   | LB           |
|----|-----------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|--------------|
| 1  | Phase 1: Zero-shot and early LoRA | LoRA Letter Log-Prob Scoring.csv                                          | r=4, q/v, 276K, 1 ep, LR=2e-5, letter-only log-prob scoring.                        | Single | 0.722        |
| 2  | Phase 1: Zero-shot and early LoRA | LoRA v2 Answer-Phrase Training Scaled Parameters.csv                      | r=16, q/k/v/o, 1.1M, 2 ep, LR=3e-5, answer-phrase training.                         | Single | 0.758        |
| 3  | Phase 1: Zero-shot and early LoRA | LoRA v2 Answer-Phrase Training Scaled Parameters_IMPROVEDZeroTraining.csv | Improved zero-training/inference variant of LoRA v2 answer-phrase setup.            | Single | 0.758        |
| 4  | Phase 1: Zero-shot and early LoRA | LoRA v3 Full-Layer Adaptation (Attention MLP) .csv                        | Attention + MLP adaptation; standard LoRA.                                          | Single | 0.754        |
| 5  | Phase 1: Zero-shot and early LoRA | LoRA v4 Balanced MLP Attention Single Epoch.csv                           | Balanced MLP + attention, single epoch.                                             | Single | 0.730        |
| 6  | Phase 1: Zero-shot and early LoRA | LoRA v5 corrected answer-token loss v2 config full validation.csv         | Corrected answer-token loss with v2 configuration and full validation.              | Single | 0.754        |
| 7  | Phase 1: Zero-shot and early LoRA | Lora v5 - inference improvement - v1 - Generation.csv                     | v5 adapter with generation-style inference improvement.                             | Single | 0.752        |
| 8  | Phase 2: Inference optimization   | LoRA v2 Inference Ablation Study.csv                                      | v2 model with native/full-prompt inference ablation; strongest v2 inference result. | Single | 0.768        |
| 9  | Phase 2: Inference optimization   | LoRA v2 Confidence-Based Inference Router.csv                             | Confidence-based routing over v2 inference variants.                                | Single | 0.758        |
| 10 | Phase 2: Inference optimization   | LoRA v6 HighRes512 Training.csv                                           | High-resolution 512 training variant.                                               | Single | 0.762        |
| 11 | Phase 2: Inference optimization   | LoRA v7 Clean Chat-Template Training.csv                                  | Chat-template/SFTTrainer-style training; masking failure.                           | Single | 0.499        |
| 12 | Phase 2: Inference optimization   | LoRA v2 Hard Sample Mining Mixed Fine-tuning.csv                          | Hard-sample mining with mixed easy/hard fine-tuning.                                | Single | 0.754        |
| 13 | Phase 2: Inference optimization   | LoRA v2 Hard Sample Mining Mixed Fine-tuning_v2.csv                       | Second hard-sample mixed fine-tuning variant.                                       | Single | 0.756        |
| 14 | Phase 2: Inference optimization   | LoRA_v2_TTA.csv                                                           | v2 test-time augmentation / choice permutation attempt.                             | Single | 0.754        |
| 15 | Phase 2: Inference optimization   | LoRA-v2-384cap-FullPrompt.csv                                             | v2 inference with 384-capped images and full prompt.                                | Single | 0.762        |
| 16 | Phase 2: Inference optimization   | LoRA-v2-NativeRes-Train.csv                                               | Native-resolution training variant.                                                 | Single | 0.752        |
| 17 | Phase 3: Advanced configs         | submission_seed42.csv                                                     | v2 reproduction baseline.                                                           | Single | 0.712        |
| 18 | Phase 3: Advanced configs         | LoRA_v9_TextOnly_NoSplit.csv                                              | r=8, all text projections, no image splitting.                                      | Single | 0.766        |
| 19 | Phase 3: Advanced configs         | ZeroShot_Correct_Baseline.csv                                             | Corrected zero-shot baseline.                                                       | Single | 0.698        |
| 20 | Phase 3: Advanced configs         | LoRA-v9-r8-all7-flatLR-trainval.csv                                       | r=8, all 7 projections, flat LR=2e-4, 3 epochs, letter-only target, train+val.      | Single | 0.861        |
| 21 | Phase 3: Advanced configs         | LoRA-v9-Inference-MetadataPrompt.csv                                      | v9 adapter with metadata prompt at inference.                                       | Single | 0.853        |
| 22 | Phase 4a: Individual models       | sub_A_cap_meta_dora_r8a16_s42.csv                                         | Model A: DoRA r=8/alpha=16, self-captioning + metadata, seed 42.                    | Single | 0.871        |
| 23 | Phase 4a: Individual models       | sub_B_cap_meta_dora_r8a16_s123.csv                                        | Model B: DoRA r=8/alpha=16, self-captioning + metadata, seed 123.                   | Single | 0.845        |
| 24 | Phase 4a: Individual models       | sub_C_cap_sol_dora_r8a16_s456.csv                                         | Model C: DoRA r=8/alpha=16, self-captioning + solution field, seed 456.             | Single | 0.787        |
| 25 | Phase 4a: Individual models       | sub_D_cap_meta_dora_r8a32_s789.csv                                        | Model D: DoRA r=8/alpha=32, self-captioning + metadata, seed 789.                   | Single | 0.863        |
| 26 | Phase 4a: Individual models       | sub_E_nocap_lora_r8a16_s42.csv                                            | Model E: LoRA r=8/alpha=16, no self-captioning, seed 42.                            | Single | <b>0.879</b> |

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Table 10: All submitted models (continued).

| #  | Phase               | Submission filename       | Key Config / Approach                             | Type     | LB           |
|----|---------------------|---------------------------|---------------------------------------------------|----------|--------------|
| 27 | Phase 4b: Ensembles | sub_ensemble_A_E_only.csv | 2-model score-average ensemble: A + E.            | Ensemble | 0.873        |
| 28 | Phase 4b: Ensembles | sub_ensemble_A_B_E.csv    | 3-model score-average ensemble: A + B + E.        | Ensemble | 0.875        |
| 29 | Phase 4b: Ensembles | sub_ensemble_no_C.csv     | 4-model score-average ensemble excluding Model C. | Ensemble | 0.885        |
| 30 | Phase 4b: Ensembles | sub_ensemble_all_noC.csv  | Alternative all-model/no-C ensemble combination.  | Ensemble | 0.883        |
| 31 | Phase 4b: Ensembles | sub_ensemble_all_FG.csv   | Ensemble with additional F/G models.              | Ensemble | 0.883        |
| 32 | Phase 4b: Ensembles | sub_ABCDEFGH_finals.csv   | Full A–H score-average ensemble.                  | Ensemble | 0.887        |
| 33 | Phase 4b: Ensembles | sub_score_average.csv     | Best score-average ensemble / final submission.   | Ensemble | <b>0.889</b> |

## 10 Results Summary

Table 11: Score progression. Total: +26.4pp from zero-shot baseline.

| Phase | Milestone                      | Type            | LB           |
|-------|--------------------------------|-----------------|--------------|
| 0     | Zero-shot baseline             | —               | 0.625        |
| 1     | LoRA v1 (letter log-prob)      | Single          | 0.722        |
| 1     | LoRA v2 (answer-phrase)        | Single          | 0.758        |
| 2     | v2 + native inference          | Single          | 0.768        |
| 3     | v9 (all 7, flat LR, train+val) | Single          | 0.861        |
| 4a    | Model E (best single)          | Single          | 0.879        |
| 4b    | 4-model ens. (no C)            | Ensemble        | 0.885        |
| 4b    | <b>Score-avg. ensemble</b>     | <b>Ensemble</b> | <b>0.889</b> |

**Cross-phase findings.** Early ablations established that MLP targets hurt under standard LoRA at  $\text{LR}=3\times 10^{-5}$ , but the v9 run later contradicted this with flat  $\text{LR}=2\times 10^{-4}$  (achieving 0.861). DoRA regularization then showed a second way to enable all-7 targets at the lower, more stable LR. This iterative progression—where each negative result motivated a revised configuration in a later phase—was central to reaching 0.889.

## 11 Negative Results

- **MLP LoRA targets (standard LoRA,  $\text{LR}=3\times 10^{-5}$ ):** Degraded across 3 experiments (0.754, 0.730, 0.766) vs. attention-only (0.758). LR-dependent: succeeded at  $\text{LR}=2\times 10^{-4}$  (v9) and with DoRA.
- **Choice-text scoring:** 0.430, far below zero-shot. Length bias.
- **SFTTrainer + chat template:** 0.499. Broken label masking, training loss=5.7.
- **Hard sample fine-tuning:** 0.754 vs. 0.768. Model fit noise, forgetting easy samples.
- **Solution field in training:** Model C scored 0.787, degraded ensembles.
- **External QLoRA (4-bit NF4):** Single attempt,  $\text{LR}=2\times 10^{-4}$ , collapsed to 0.400.
- **512×512 training:** 0.762 vs. 0.768 native inference on 224-trained model.
- **Confidence routing:** Val +0.5pp didn’t transfer to LB (0.758 vs. 0.768).
- **TTA (choice permutation):** Degraded from 0.768 to 0.754.
- **High LR ( $5\times 10^{-4}$ ):** Catastrophic forgetting, scored 0.515.
- **Chat template zero-shot:** 0.450 vs. 0.635 raw prompt.



## 12 Technical Issues and Debugging

We encountered and resolved 16 distinct technical issues across all phases.

### 12.1 Training Pipeline Bugs

1. **Gradient checkpointing crash.** `gradient_checkpointing_enable()` on a PEFT model raised `RuntimeError` because PEFT freezes base parameters, breaking gradient flow. *Fix:* Call `enable_input_require_grads()` first; use `use_reentrant=False`.
2. **LR destroying pretrained knowledge.**  $LR=5\times 10^{-4}$  scored 0.515—below zero-shot 0.625. *Fix:*  $LR=3\times 10^{-5}$  ( $17\times$  lower).
3. **Image truncation at 384px.** 384px images generated  $\sim 729$  image tokens, silently truncating the answer letter from the sequence end. Collator diagnostic showed last token as “the” instead of “A”. *Fix:* 224px for training; native at inference.
4. **Batch vs. individual processor mismatch.** Batch processing misaligned label masks in  $\sim 87\%$  of samples (7/8 in diagnostic). *Fix:* Process individually in `__getitem__`, manually left-pad in collator.
5. **Label masking misalignment.** `prompt_len` from a separate processor call didn’t account for variable image token counts. *Fix:* Tokenize answer separately, unmask last  $N$  tokens from sequence end.
6. **Single-token loss instability.** Loss fluctuated 0.001–2.5 per step, running average appeared increasing even during learning. *Fix:* Answer-phrase training (5–15 tokens of supervision).
7. **LoRA params exceeded 5M.**  $r=16$  on all 7 projections yielded 9.57M parameters (nearly  $2\times$  limit). *Fix:*  $r=8$  (4.78M) or  $r=16$  attention-only (1.1M).
8. **PeftModel loaded with frozen params.** `from_pretrained()` loaded with `requires_grad=False`; optimizer got empty param list, raising `ValueError`. *Fix:* Manually enable gradients for “lora” parameters.

### 12.2 Inference and Environment Bugs

9. **Right-padding corrupted generation.** Default right-padding breaks decoder-only generation. *Fix:* `padding_side='left'` for inference.
10. **GPU OOM during validation.** Batch size 16 after training needed 6.38 GiB with only 1.91 GiB free. *Fix:* `Batch 4 + empty_cache() + gc.collect()`.
11. **Pixel tensor stacking failure.** Native-resolution images have variable sizes; `torch.stack` fails with shape mismatch. *Fix:* Batch size 1 for native training.
12. **RAM crash during pre-tokenization.** Crashed at 62% (194/311 batches); `pixel_values`  $\sim 3$  MB each overwhelmed Colab RAM. *Fix:* On-the-fly processing in `__getitem__`.
13. **Colab runtime disconnections.** Runs exceeding 2.5 hours lost all progress. *Fix:* Checkpoint to Google Drive every 500 steps.
14. **Transformers import error.** `AutoModelForVision2Seq` unavailable after package upgrade. *Fix:* `Pin transformers==4.57.1`.
15. **WandB blocking training.** Interactive prompt halted `SFTTrainer` execution. *Fix:* `WANDB_DISABLED=true` environment variable.
16. **SFTTrainer API change.** `tokenizer` parameter renamed to `processing_class` in newer TRL versions. *Fix:* Use updated parameter name.

## 13 Team Contributions

Ananya Shashidhara Nadig primarily focused on systematic ablation of LoRA targets, ranks, label masking, image resolution, prompt content, and inference methods (experiments 1–21 in Table 10). This included 26 controlled experiments, identification and resolution of all 16 technical bugs, hard sample analysis, confidence analysis, and inference optimization (native resolution, TTA, confidence routing, multi-prompt ensemble).



**Krishan Kumar Gupta** primarily focused on building on WS1 findings to develop the final pipeline including self-captioning, metadata enrichment, DoRA adaptation, multi-seed training, and score-average ensembling (experiments 22–33). This included training 8 diverse model configurations (Models A–H) and systematic ensemble composition ablation.

Both teammates contributed to prompt design, validation analysis, and report writing. The final pipeline integrated insights from both workstreams.

## 14 Reproducibility

### 14.1 Dependencies

```
transformers==4.57.1
peft==0.18.0
bitsandbytes>=0.45.0
accelerate>=1.7.0
pandas==2.2.2
Pillow==11.3.0
tqdm
torch>=2.0
kagglehub
```

### 14.2 Repository

The full project repository is available at <https://github.com/Krishan101/kaggle-dl-final-competition>. The repository includes a `requirements.txt` file and all training/inference notebooks.

### 14.3 Reproduction Steps

1. **Environment:** Clone the repository and install dependencies from `requirements.txt`.
2. **Data:** Download competition data via `kagglehub.competition_download('pixels-to-predictions')`.
3. **Self-captioning:** Run `self_captioning.ipynb` to cache image descriptions.
4. **Training:** Run `phase2_lora_v2_train.ipynb` (v2 baseline), `lora_v9_train.ipynb` (v9), and `phase4_dora_train.ipynb` (DoRA/ensemble models A–H).
5. **Inference:** Run `phase4_ensemble_inference.ipynb` to load checkpoints, perform native-resolution log-likelihood scoring, and generate score-average ensemble.
6. **Submission:** Output `submission.csv` with columns `id` and `answer` (0-indexed integer).

## 15 Conclusion

Across 33 leaderboard submissions and 16 bug fixes, we improved from a zero-shot baseline of 0.625 to a final score of **0.889** on the ScienceQA visual multiple-choice task. Key findings:

1. **LoRA target–LR interaction:** MLP targets hurt at  $\text{LR}=3\times 10^{-5}$  under standard LoRA but succeed at  $\text{LR}=2\times 10^{-4}$  (v9) or with DoRA regularization.
2. **DoRA provided a stable alternative for broader targeting** at stable learning rates via magnitude-direction decomposition.
3. **Self-captioning has modest individual impact** but increases model diversity, amplifying ensemble gains.
4. **Score-average ensembling** across diverse configurations yields +1.0pp over the best individual model, with diminishing returns from weaker models.
5. **Native image resolution** at inference preserves diagram details (+1.0pp), outperforming higher-resolution training.
6. **Lecture/hint context** contributes 7–12pp; removing either degrades substantially.

7. **Answer-phrase training** provides  $5\text{--}15\times$  more gradient signal than letter-only (+3.6pp).
8. **Solution field training hurts:** Including reasoning explanations degrades accuracy and ensemble quality.

## Limitations

Several limitations affect our work. First, some early ablations (experiments 1–2) confounded rank, alpha, and target module changes simultaneously; we note these explicitly in §7.1. Second, the 200-sample validation subset used in early experiments was unreliable; we switched to the full validation set for all later decisions. Third, self-captioning was limited by the base model’s weak vision encoder. Fourth, QLoRA was attempted only once with an excessively high LR, leaving this direction underexplored. Fifth, while the v9 run’s flat  $\text{LR}=2\times 10^{-4}$  achieved strong results, we did not fully explore its interaction with DoRA or epoch count.

## AI Tools Disclosure

We used the following AI assistants during this project:

- **Claude (Anthropic):** Code generation, debugging assistance, training pipeline development, inference optimization, and report writing assistance.
- **ChatGPT (OpenAI):** Strategy development, ablation planning, and alternative approach exploration.

Both tools were used for coding and debugging support, not for generating competition answers. All model outputs were produced by our fine-tuned models.

## References

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